Interpretable third-order penalized trend estimation for vehicle jerk: relative-noise calibration and forecasting

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ARTICLE INFO

Keywords:
Penalized least squares
third differences
jerk
smoothness index
relative spectral noise
calibration
NGSIM

ABSTRACT

Estimating physical jerk from vehicle positions is difficult because third differences amplify observation noise. We investigate third-order penalized least-squares (PLS) trend estimation with a nonzero reference third difference and an interpretable normalized smoothness index S. We describe the relationship between S, effective degrees of freedom and, under an independent Gaussian variance-ratio model, relative spectral attribution of third-difference variability to observation noise. Two simulation studies supply known position and jerk: a 1,296-trajectory comparison of PLS smoothing choices with established spline and local-polynomial filters, and a 1,944-condition paired calibration study with 72 independent random-innovation parents. In the correctly specified Gaussian calibration setting, joint restricted-likelihood selection achieves Spearman association 0.965 between estimated and theoretical S (mean absolute error 0.0115); serially correlated observation errors substantially degrade calibration. In the known-jerk comparison, a global cubic reference mean with profiled Gaussian likelihood performs well for jerk recovery, whereas no smoothing rule is uniformly best for subsequent position forecasts. We then evaluate recorded-future position predictions on 200 real NGSIM trajectory episodes and compare PLS with GCV smoothing splines, Savitzky–Golay and polynomial baselines; the spline has the lowest pooled forecast error at all evaluated horizons. Third-order PLS offers a physically meaningful smoothing coordinate, not a universally optimal forecast or an absolute measure of sensor precision.

1. Introduction

Vehicle trajectories provide a physical interpretation for higher-order trend smoothing: the first three time derivatives of longitudinal position are velocity, acceleration and jerk. The US Federal Highway Administration's Next Generation Simulation (NGSIM) dataset reports camera-derived vehicle positions every 0.1 seconds (U.S. Department of Transportation, Federal Highway Administration, 2016). Earlier vehicle-trajectory research emphasizes that estimating acceleration from these observations magnifies measurement noise (Thiemann, Treiber and Kesting, 2008); third derivatives amplify it further. This motivates examining a third-difference penalty ($d = 3$), rather than restricting the study to the commonly used second-difference penalty ($d = 2$).

We study the PLS model of Guerrero (2007), its trace-based smoothness index (Guerrero, 2008) and classical tuning criteria (Cortés-Toto, Guerrero and Reyes, 2017). The statistical questions are whether a third-order fit recovers independently known jerk, whether normalized smoothness quantifies relative spectral observation noise under a specified stochastic model, and how the fitted trends compare with conventional smoothers in forecasting recorded positions. We answer them with two distinct synthetic studies followed by a larger real-data case study. Numerical solver timing and matrix-factorization benchmarks are outside this paper's scope.

We also compare two boundary continuation rules for the same fitted $d = 3$ PLS trend and benchmark them against natural cubic smoothing splines with GCV-selected penalties, Savitzky–Golay differentiation and local polynomial fits. These are standard local and global approaches to derivative estimation from noisy experimental data (Ahnert and Abel, 2007); their smoothing penalties and boundary prediction rules are not interchangeable with those of third-difference PLS. The resulting analysis tests an implementation and application of an established PLS model, not an assertion of historically first use of $d = 3$ differences.

The real NGSIM dataset supplies camera-derived position, but not independently measured latent truth or jerk. We do not contaminate the real positions to create artificial validation labels. In the simulation studies we first evaluate 1,296 known-jerk analytic trajectories under different motion/noise regimes, and then independently vary observation-noise scale and genuine jerk volatility across 1,944 paired factorial conditions (72 independent random-draw parents). Only these known-truth studies support direct calibration and physical-jerk error assessment. The primary NGSIM evaluation instead uses 200 recorded episodes from two freeway sites.

1.1. The noise amplification motivating third-order smoothing

If measurement noise ε_t has independent variance σ_ε^2, third forward differentiation uses stencil $(-1, 3, -3, 1)$, hence

$$\mathrm{Var}\left(\frac{D_3\varepsilon_t}{(\Delta t)^3}\right) = \frac{20\sigma_\varepsilon^2}{(\Delta t)^6}. \tag{1}$$

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With $\Delta t = 0.1$ s, direct third differences can be extremely unstable. Equation (1) is a conditional mathematical illustration, not an empirical noise-variance estimate: camera-tracking errors may be temporally correlated. Regularization suppresses variation but can bias sudden real changes in jerk. Thus plotted derivatives remain *estimates*, not ground truth.

2. Third-order penalized least-squares trend estimation

2.1. Trend and reference third differences

Let $y = (y_1, \dots, y_n)^\top$ denote equally spaced longitudinal positions sampled at interval Δt. Write τ for the unobserved positional trend. The order-d difference matrix $D_d \in \mathbb{R}^{(n-d)\times n}$ annihilates polynomials of degree at most $d-1$. The penalized least-squares (PLS) estimator of Guerrero (2007) is

$$\widehat{\tau}_{\lambda,\mu} = \arg\min_{\tau\in\mathbb{R}^n} \left\{ \|y-\tau\|_2^2 + \lambda \|D_d\tau - \mu\mathbf{1}\|_2^2 \right\}, \qquad (2)$$

where $\lambda \geq 0$ controls regularization and μ is the reference mean of the dth differences. For $Q_d = D_d^\top D_d$, differentiation gives the unique solution at finite λ:

$$\widehat{\tau}_{\lambda,\mu} = (I + \lambda Q_d)^{-1}(y + \lambda\mu D_d^\top \mathbf{1}). \qquad (3)$$

The estimator is historical trend reconstruction; forecasting requires an additional extension rule.

We focus on $d = 3$ because $D_3\tau_i/(\Delta t)^3$ is a discrete approximation to physical jerk, the time derivative of acceleration. When $\mu = 0$, strong penalization favors a quadratic position trend (approximately constant acceleration). For fixed nonzero μ, the limiting trend is cubic with prescribed discrete third difference, representing approximately constant nonzero jerk. These are *modeling assumptions*, not independent physical measurements. Results for other difference orders provide secondary comparisons, not the central claim.

2.2. An interpretable smoothing coordinate

Let $\delta_1, \dots, \delta_{n-d} > 0$ be the nonzero eigenvalues of Q_d; the remaining d eigenvalues are zero because $\dim\ker D_d = d$. Define the normalized smoothness index

$$S_d(\lambda) = \frac{1}{n-d}\sum_{j=1}^{n-d} \frac{\lambda\delta_j}{1+\lambda\delta_j} \in [0,1]. \qquad (4)$$

For fixed d and n, this is monotone in λ, with $S_d(0) = 0$ and $S_d(\infty) = 1$. Thus S provides a comparable, bounded description of imposed trend smoothness rather than requiring the reader to interpret arbitrary penalty magnitudes. For *fixed* μ, the smoothing matrix is $H_\lambda = (I + \lambda Q_d)^{-1}$ and its effective degrees of freedom satisfy

$$\mathrm{df}(H_\lambda) = \mathrm{tr}(H_\lambda) = n - (n-d)S_d(\lambda). \qquad (5)$$

For $d = 3$, $n = 80$, this varies from 80 to 3. If μ is estimated from the same observations, the effective complexity of the *complete* fitted estimator may be larger, so Eq. (5) should not be interpreted as its unconditional degrees of freedom.

The stochastic variance-ratio interpretation of Guerrero (2007) and the percentage-smoothness work of Guerrero (2008) motivate this index. Our contribution is not to introduce PLS or the original notion of smoothness, but to examine S for third-order vehicle kinematics with known-truth calibration and observed trajectories. Section 4 tests the stronger *model-conditional* spectral noise interpretation; the index alone does not measure absolute sensor precision.

2.3. Selecting λ and the reference μ

We distinguish the *reference-mean policy* from the *smoothing criterion*. Reference means are fixed $\mu = 0$, the observed training mean $\widehat{\mu}_{\mathrm{raw}} = \mathbf{1}^\top D_3 y/(n-3)$, and a cubic OLS reference $\widehat{\mu}_{\mathrm{OLS}} = 6\widehat{a}_3(\Delta t)^3$ when a cubic in physical time has coefficient $\widehat{a}_3$. The raw difference mean is sensitive to measurement errors at the boundary. Our synthetic-only known-μ method is explicitly an oracle *control*, never a practical estimator on real observations.

For the same fixed-μ PLS fit, historical training observations select λ through generalized cross-validation (GCV) or a profiled Gaussian variance-components likelihood. Classical ordinary leave-one-out CV, AICc and a constrained residual BIC are also described for the preliminary real pilot, following Cortés-Toto et al. (2017); those pilot criteria are not five independent pieces of forecast evidence. In historical experiments the search is over predeclared candidate S values; the later likelihood comparisons refine a continuous one-dimensional search and explicitly check the variance-ratio boundary limits. Grid and continuous results are identified separately.

For the Gaussian model

$$y = \tau + \varepsilon, \quad D_3\tau = \mu\mathbf{1} + u, \quad \varepsilon \sim N(0, \sigma_\varepsilon^2 I), \quad u \sim N(0, \sigma_u^2 I), \qquad (6)$$

with independent disturbances, set $B = D_3 D_3^\top$, $m = n-3$ and $W_\lambda = I_m + \lambda B$. The variance ratio $\lambda = \sigma_\varepsilon^2/\sigma_u^2$ defines the fixed-μ Gaussian restricted likelihood of $D_3 y - \mu\mathbf{1}$. When μ is unknown, the generalized least-squares mean at each penalty is

$$\widehat{\mu}(\lambda) = \frac{\mathbf{1}^\top W_\lambda^{-1} D_3 y}{\mathbf{1}^\top W_\lambda^{-1}\mathbf{1}}. \qquad (7)$$

Writing $r_\lambda = D_3 y - \widehat{\mu}(\lambda)\mathbf{1}$, the corresponding mean-adjusted profiled restricted likelihood, apart from penalty-independent terms, is

$$F_R(\lambda) = \log|W_\lambda| + \log(\mathbf{1}^\top W_\lambda^{-1}\mathbf{1}) + (m-1)\log\left(\frac{r_\lambda^\top W_\lambda^{-1} r_\lambda}{m-1}\right). \qquad (8)$$

This Gaussian extension to Guerrero's original second-moment formulation is statistically distinct from simply inserting a sample-derived μ in the *fixed*-mean likelihood. It does not guarantee minimum physical jerk or forecast error.

2.4. Conventional smoothers and comparable outcomes

To evaluate PLS against established derivative estimators, the simulation and real-data studies include natural cubic smoothing splines with a training-only GCV-selected second-derivative penalty (Craven and Wahba, 1979), Savitzky–Golay local polynomial filtering (Savitzky and Golay, 1964), and local quadratic or cubic polynomial trends. For prediction, simple constant-velocity and constant-acceleration extrapolations provide additional physical baselines. These methods are distinguished from PLS by both their regularization constraints and their extrapolation rules; in particular, a natural cubic smoothing spline penalizes a *second* derivative, whereas D_3 PLS penalizes discrete *third* differences. We do not assert that one parameterization of smoothness makes these estimators identical (Ahnert and Abel, 2007).

The PLS historical fit is extrapolated either by a local polynomial fitted to the final 25 smoothed observations or by the recurrence obtained from the last three fitted values. With ∇ denoting a backward difference, the latter is

$$\hat{\tau}_N(h) = \hat{\tau}_N + h\nabla\hat{\tau}_N + \binom{h+1}{2}\nabla^2\hat{\tau}_N + \binom{h+2}{3}\mu, \qquad h \geq 1. \tag{9}$$

It is the direct $d = 3$ algebraic extension of the constant-difference extrapolation logic discussed by Guerrero (2007) for lower orders, *not* a separate optimal probabilistic forecast model. Natural cubic smoothing-spline forecasts use training-fitted endpoint derivatives or a specified local polynomial continuation. Every reported forecast is scored on chronologically withheld future positions, with no future observations used to choose the smoothing or reference parameters.

We report jerk RMSE only where a synthetic generating trajectory supplies the true discrete third difference. On camera-derived NGSIM positions, the target is held-out *recorded position*; physically true jerk and observation error are unavailable.

3. Simulation study I: recovering known jerk and forecasting clean position

3.1. Analytic trajectories and conventional comparators

A simulation experiment can evaluate the accuracy of jerk reconstruction because the generative position and its third difference are both known. We use the previously completed Experiment005, which specifies six analytic motion regimes, three measurement-noise magnitudes and three observation-error structures. Its independent confirmation stage contains 1,296 seeded trajectories, with a separate 96-trajectory pilot. Each trial provides 80 historical positions at $\Delta t = 0.1$ s and 20 subsequent clean positions for forecast evaluation. The true reference jerk is the *discrete* quantity

$$j_i^{\text{true}} = \frac{D_3\tau_i^{\text{true}}}{(\Delta t)^3},$$

rather than an exact pointwise derivative improperly substituted across motion regime changes.

We compare third-order PLS with several training-only reference policies and smoothing criteria, as well as natural cubic smoothing splines with GCV and Savitzky–Golay local polynomial filtering. We evaluate the historical jerk reconstruction error against known truth, and separately evaluate the *clean future position* withheld from every fitting and selection procedure. The oracle reference mean uses known historical truth for a synthetic-only control and is unavailable in real data. No method receives its future observations.

For a fitted trend $\hat{\tau}$, the historical mean squared jerk error is

$$\text{MSE}_j = \frac{1}{n-3}\sum_{i=1}^{n-3}\left[\frac{D_3\hat{\tau}_i}{(\Delta t)^3} - j_i^{\text{true}}\right]^2. \tag{10}$$

We report pooled RMSE in m/s^3. Forecast errors, in contrast, have position units of meters, and are not combined with jerk losses.

Table 1 illustrates a substantial distinction between training-based GCV and Gaussian profile-likelihood selection for estimating third differences under these generative noise assumptions. A globally fitted cubic reference also avoids the endpoint-noise sensitivity of the raw mean of third differences. It would be incorrect to conclude that PLS always dominates splines: the covariance model, target derivative order, and selection rule differ between methods, and the real-data comparison in Section 5 gives a different forecast ranking.

3.2. Joint reference-mean estimation and future position

Experiment006 reanalyses *exactly* the same 1,296 archived confirmation trajectories, without generating new independent noise realizations. It compares a continuous search for the Gaussian penalty with either zero μ, a sample or global cubic OLS reference, and the joint unknown-mean REML estimator of Eq. (8). A matched 15-candidate joint-REML selection separates the choice of search domain from mean estimation. Additional reference shrinkage is selected by rolling-origin validation *inside* each historical training window.

The better reference mean for *historical jerk* need not be the one that forecasts clean future position best. For example, the global cubic OLS/profile-likelihood policy has the smallest historical jerk RMSE among the tabulated PLS variants (2.4413 m/s^3), whereas the fixed-zero reference has the smallest one-second clean position RMSE of these variants (0.4053 m). Properly adjusted joint REML is statistically meaningful, but its jerk error and forecast error are not smaller than the alternatives in this designed simulation.

Table 1
Experiment005: pooled historical known discrete-jerk RMSE (m/s^3) in 1,296 independently generated analytic confirmation trajectories. These methods share the frozen simulation and training-only smoothing selection. PLS profile ML uses the historical 15-value smoothness grid; the cubic OLS reference is estimated from noisy training positions. The oracle uses known truth and is not available in applications. These are results under the specified motion/noise regimes, not universal rankings of derivative estimators.

Historical jerk estimator	RMSE (m/s^3)
PLS $d = 3$, global cubic OLS $\widehat{\mu}$, profile ML	**2.9177**
PLS $d = 3$, known oracle μ, profile ML	2.9182
PLS $d = 3$, fixed $\mu = 0$, profile ML	3.0207
PLS $d = 3$, raw sample $\widehat{\mu}$, profile ML	9.4957
Savitzky–Golay	36.8291
GCV-selected natural cubic smoothing spline	60.9131
PLS $d = 3$, fixed $\mu = 0$, GCV	68.2422

Table 2
Experiment006 continuous/profile and joint-REML reanalysis of the *same* 1,296 known-truth simulation trials as Table 1. Historical known-jerk RMSE (m/s^3) and one-second clean future-position RMSE (meters) are different estimands. Position forecast RMSE uses the local-25 continuation; the historical jerk RMSE does not. The fixed-mean profile scores condition on the selected reference, whereas joint REML estimates that reference at each penalty.

Smoothing and reference policy	Jerk RMSE	Future position RMSE
PLS, $\mu = 0$, continuous profile ML	2.5874	**0.4053**
PLS, cubic OLS $\widehat{\mu}$, continuous profile ML	**2.4413**	0.4353
PLS, unknown-μ joint REML, continuous	3.2950	0.4977
PLS, unknown-μ joint REML, 15-S grid	4.1014	0.4966
PLS, sample $\widehat{\mu}$, continuous profile ML	7.7077	–
Natural cubic smoothing spline, GCV + quadratic Taylor	–	0.9710

Figure 1 visualizes these distinct targets and also previews the observed-position NGSIM result discussed later. The figure does not pool measurements across incompatible units.

The following simulation study asks a separate question: does the index S measure a known *relative noise fraction* when observation noise and genuine jerk variability are manipulated independently? This interpretation cannot be assessed from forecast RMSE or from visually smooth jerk curves alone.

4. Simulation study II: calibration of the smoothness index

4.1. What the normalized index identifies

For a third-order difference operator $D = D_3$ and a fixed reference μ, the model $y = \tau + \varepsilon$, $D\tau = \mu\mathbf{1} + u$, with independent Gaussian disturbances $\varepsilon \sim N(0, \sigma_\varepsilon^2 I)$ and $u \sim N(0, \sigma_u^2 I)$, implies

$$\lambda = \frac{\sigma_\varepsilon^2}{\sigma_u^2}, \quad S(\lambda) = \frac{1}{m}\sum_{j=1}^{m} \frac{\lambda\delta_j}{1+\lambda\delta_j}, \quad m = n-3. \tag{11}$$

On the spectral coordinates of Dy, the jth summand is the *fraction of variance due to differenced observation error*, not the fraction of observations that are erroneous. Figure 2 illustrates these spectral weights and the exact conditional complexity identity $\mathrm{df}(H_\lambda) = n - (n-3)S(\lambda)$. This latter identity holds for a *fixed* reference mean; if μ is estimated, the unpenalized effective complexity can increase.

If observation noise and genuine jerk innovations are correlated in time, the scalar variance-ratio formula no longer gives the correct covariance allocation. Nevertheless, the known generative covariance matrices define a target for validation:

$$S_{\mathrm{cov}} = \frac{1}{n-3}\,\mathrm{tr}\left[(C_u + C_\varepsilon)^{-1} C_\varepsilon\right], \tag{12}$$

where $C_u = \mathrm{Cov}(D\tau - \mu\mathbf{1})$ and $C_\varepsilon = \mathrm{Cov}(D\varepsilon)$. Under independent Gaussian disturbances Eq. (12) equals Eq. (11) at the true variance ratio. Under covariance misspecification an estimated S need not estimate S_{cov}.

4.2. A known-jerk experiment with paired noise interventions

We independently manipulate position-measurement noise and the actual fluctuations of physical jerk. Each of $n = 80$ clean positions is constructed from a prescribed true third difference $D_3\tau = (\Delta t)^3 j$. Experimental levels are three observation-noise standard deviations $(0.01, 0.05, 0.20\,\mathrm{m})$ and three jerk-innovation standard deviations $(0.5, 2, 8\,\mathrm{m/s^3})$. We cross these with independent, AR(1), and terminal-contaminated measurement errors, and with independent, AR(1), and abrupt-reversal jerk processes. The terminal outlier has randomized sign, so its mixture covariance is

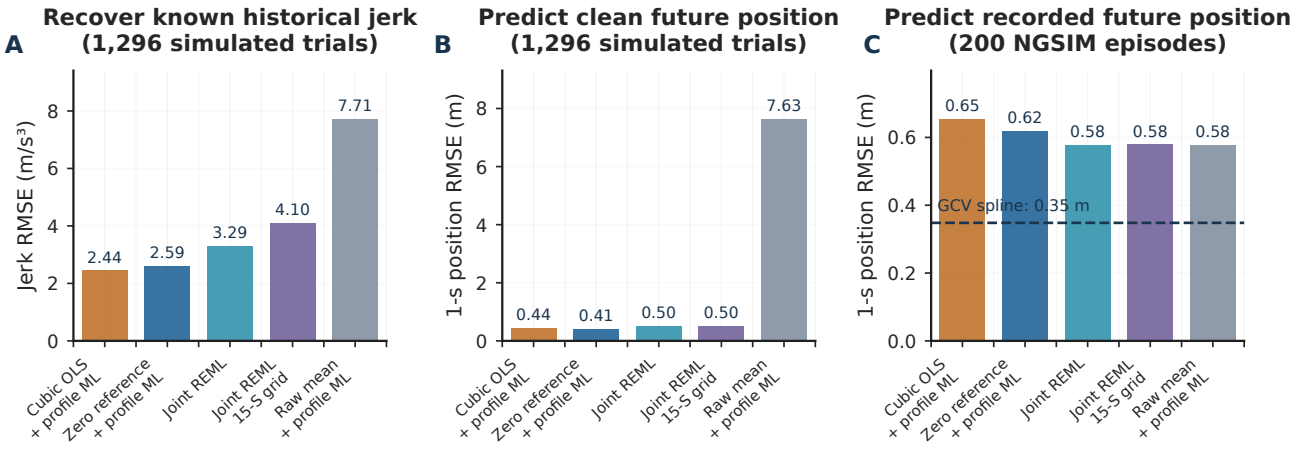


Figure 1: Estimator choice depends on the scientific outcome. (A) Known historical discrete jerk RMSE on the archived 1,296-trajectory simulation. (B) One-second forecast RMSE for *clean simulated future position*. (C) One-second forecast RMSE for *recorded real NGSIM future position* over 200 episodes, with GCV smoothing spline reference marked. Panels are different estimands with different units and source populations; they are not one pooled ranking.

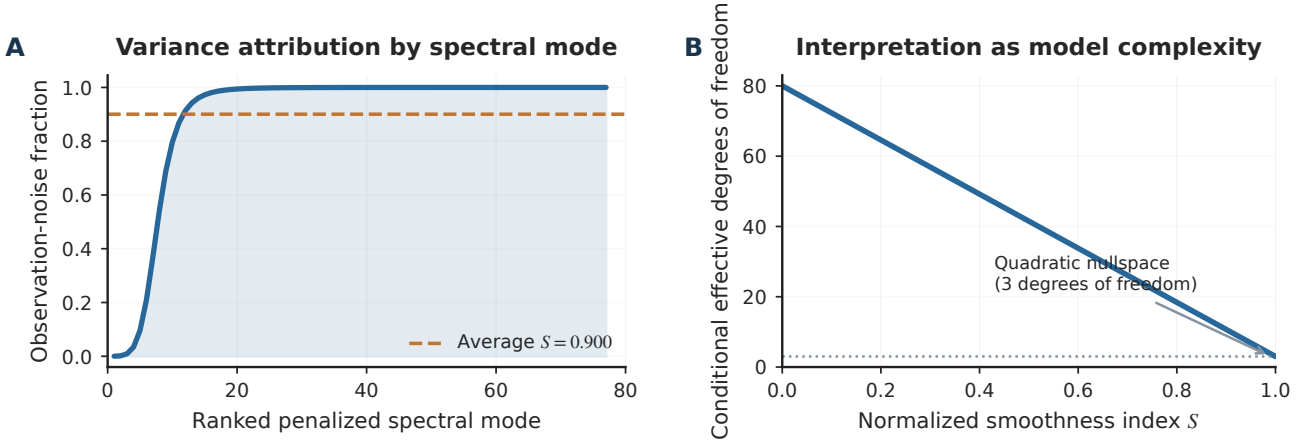


Figure 2: Meaning of the normalized $d = 3$ smoothness index. (A) Ranked observation-noise variance fractions in 77 penalized third-difference spectral directions for $n = 80$, $\Delta t = 0.1$ s, observation SD 0.05 m and true jerk innovation SD 2 m/s^3 under the independent Gaussian model; their average is S. (B) Exact conditional effective degrees of freedom as a function of S when the mean of differences is fixed. Neither panel treats S as an absolute sensor-noise standard deviation.

known, but the observations are not Gaussian. The reversal violates the constant jerk-reference assumption.

The frozen confirmatory design has 1,944 factorial cases and four smoothing selectors per case. *Only 72 independent parent random draws underlie the 1,944 cases*: standardized jerk innovations and measurement-error draws are held constant across all 27 treatments within each parent. The 162-case pilot uses a separate seed namespace. All 7,776 confirmatory fits were valid. The four selectors are Gaussian profile likelihood with the known generative μ (a simulation-only control), unknown-mean Gaussian REML, conditional profile likelihood with noisy global cubic OLS $\hat{\mu}$, and zero-mean GCV on the original 15-point smoothness grid.

4.3. Calibration and independent changes in noise and jerk

Under the correctly specified independent Gaussian model, unknown-mean REML produced Spearman correlation 0.965 between selected S and the true spectral noise fraction, with absolute calibration error 0.0115 averaged over the 216 paired factor combinations (24 stochastic parents). The known-reference control achieved 0.967 and 0.0107, respectively, while the global cubic OLS plug-in achieved 0.966 and 0.0143. The zero-reference GCV comparator achieved 0.901 and 0.0249. These are *descriptive* correlations over dependent factorial cases, not independent sample significance tests.

For unknown-mean REML under independent Gaussian errors, raising position-noise SD from 0.01 to 0.20 m at fixed jerk SD increased fitted S by 0.1498 (known theoretical

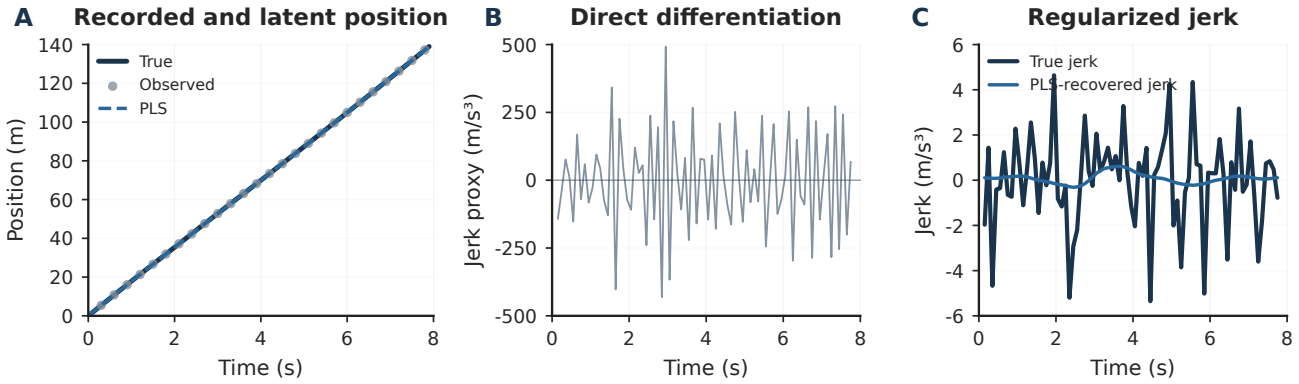


Figure 3: One *predeclared* analytic trajectory from the independent Gaussian regime (confirmation trial 000010; observation SD 0.05 m, jerk-innovation SD 2 m/s^3). (A) Prescribed clean position, noisy observations and PLS fit using global cubic OLS $\hat{\mu}$ and profiled Gaussian likelihood; (B) the raw third difference divided by $(0.1\,\mathrm{s})^3$, with severe noise amplification; (C) known prescribed jerk and the PLS-reconstructed third difference. The raw and regularized jerk panels have different vertical scales. This generated example is *not* a claimed independent real NGSIM jerk measurement.

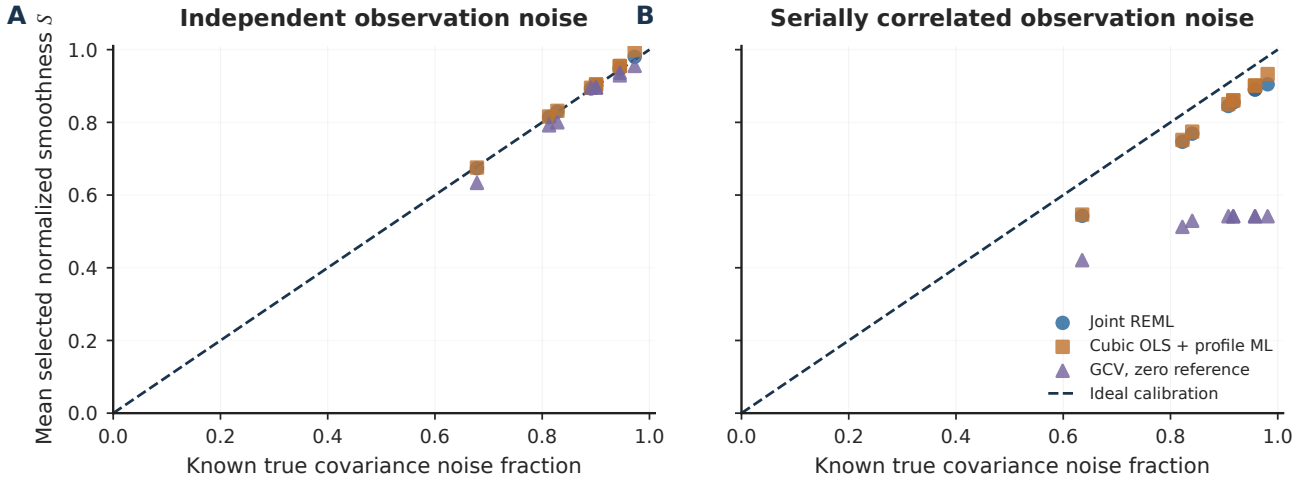


Figure 4: Empirical calibration of the selected smoothness coordinate against the *true covariance-based* noise fraction. Each point aggregates 24 independent parents at one of nine observation-noise/jerk-volatility combinations; three practical selectors are shown. (A) Independent observation noise and independent jerk innovations. (B) Correlated AR(1) observation noise with otherwise independent jerk innovations: rank association persists, but calibrated values systematically depart from the identity line. An oracle known-μ control is excluded from this practical-selection visualization.

increase 0.1451). Raising actual jerk innovation SD from 0.5 to 8 m/s^3 at fixed observation noise *decreased* fitted S by 0.1387 (known theoretical decrease 0.1337). Paired parent-level bootstrap intervals were $[0.1454, 0.1544]$ and $[-0.1428, -0.1345]$, respectively, conditional on this particular simulation design. Figure 5 shows the corresponding results across all nine covariance and maneuver regimes, including the less accurate misspecified cases.

A direct identifiability counterexample prevents interpreting S as *absolute measurement precision*: increasing $(\sigma_\varepsilon, \sigma_j)$ from (0.05 m, 2 m/s^3) to (0.20 m, 8 m/s^3) preserves the variance ratio and therefore S under the independent Gaussian model, even though the observation-noise variance increases sixteenfold. The same-ratio experiment produced an estimated mean ΔS statistically indistinguishable from zero by design. The index is relative to dynamical innovation variation, not a stand-alone metrological rating of recording quality.

4.4. Where the precision interpretation fails

Figure 6 displays calibration errors under deliberately misspecified noise and dynamics. For joint REML, mean absolute error increased from 0.0115 under independent Gaussian disturbances to 0.0712 when the measurement error had AR(1) correlation of 0.65, and to 0.1111 when both the measurement noise and jerk innovations were serially correlated. Hence favorable rank correlation alone is insufficient to claim quantitatively accurate noise attribution.

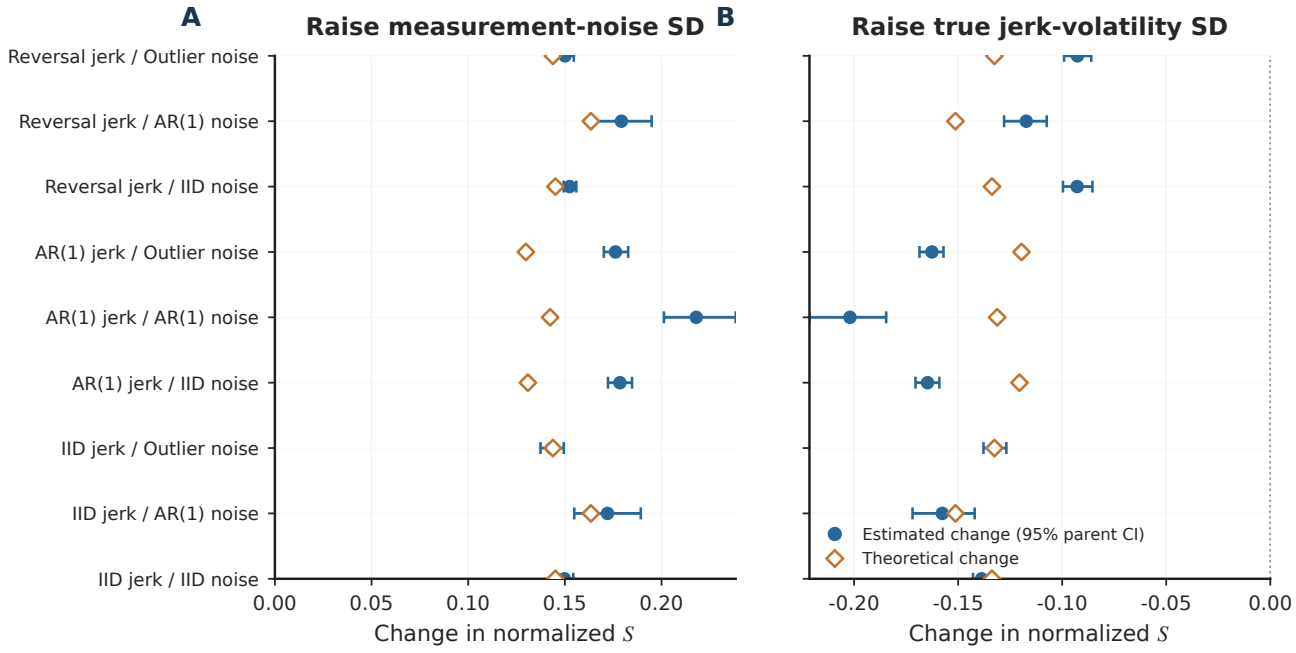


Figure 5: Two separately controlled interventions for selected joint-REML smoothness across nine jerk/measurement-error structures. (A) Increasing observation-noise SD with true jerk volatility fixed; (B) increasing true jerk volatility with observation noise fixed. Blue circles and whiskers show parent-cluster means and 95% Monte Carlo bootstrap intervals; open orange diamonds are known covariance-theoretic effects. Theoretical effects vary by correlation regime. Paired factor combinations within each independent parent are never resampled as independent observations.

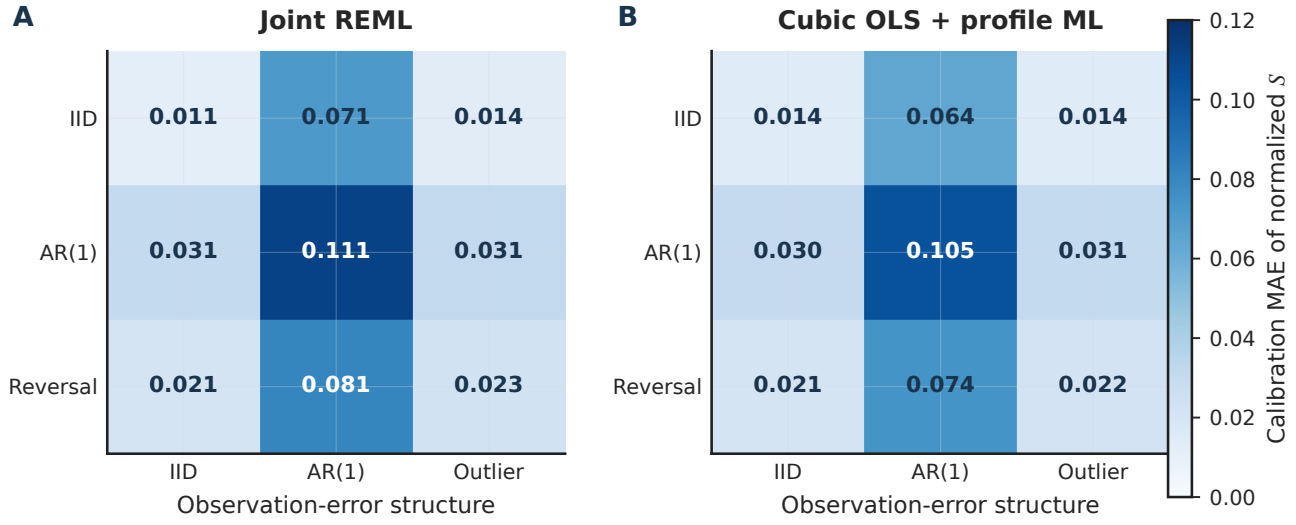


Figure 6: Absolute S calibration error against the *known* covariance-noise fraction, by true jerk structure (rows) and observation-error structure (columns). Left: properly mean-adjusted joint REML; right: global cubic OLS reference followed by fixed-mean Gaussian profile likelihood. The independently Gaussian regime occupies the upper-left cell. Darker values indicate larger errors; the AR(1) observation-error cases are conspicuous failures of the independent-noise statistical model.

4.5. Interpretation across the two simulation studies

The preceding experiments support different conclusions because they target different statistical quantities. The first simulation study finds that a global cubic OLS reference with Gaussian profile likelihood gives smaller *historical jerk-reconstruction error* than joint REML in the specified motion families (Table 2). The present controlled factorial study instead shows that joint REML can provide better *calibration of the relative spectral noise fraction* under the correctly specified independent Gaussian model, while substantially miscalibrating it for correlated observation noise.

Table 3
Main real-data comparison: pooled future *recorded-position* RMSE (meters) over 200 NGSIM episodes. GCV or conditional profile Gaussian likelihood selects smoothing using training observations only; the three-point and local-25 rules are separate continuations of a fitted PLS trend. All rows have 200 valid episodes at each horizon. Bold highlights the smallest pooled RMSE, not statistical superiority over a population.

Smoother and continuation	0.5 s	1 s	2 s
Natural cubic smoothing spline, GCV + quadratic Taylor	**0.1354**	**0.3480**	**0.8787**
PLS $d = 3$, $\mu = 0$, profile ML, three-point	0.1433	0.5124	1.8686
PLS $d = 3$, $\mu = 0$, GCV, three-point	0.1453	0.5166	1.8805
PLS $d = 3$, sample $\widehat{\mu}$, profile ML, three-point	0.1457	0.5310	2.0142
PLS $d = 3$, sample $\widehat{\mu}$, profile ML, local-25	0.3166	0.5764	1.3535
Raw local quadratic polynomial, 25 points	0.3406	0.6184	1.3447
Savitzky–Golay, quadratic Taylor	0.3547	0.8506	2.4106

The outcomes are not contradictory: minimum historical jerk loss, minimum future-position loss, and accurate noise attribution are distinct targets. The application to recorded NGSIM positions in Section 5 tests forecasting, not the unknown physical jerk or recording-sensor precision. The full simulation results, parent-level bootstrap data, generative truth and plots remain in the Experiment005–007 archives.

5. Real-data application: NGSIM vehicle trajectories

5.1. Data, observed outcomes and evaluation protocol

We use camera-derived longitudinal positions from US-101 and I-80 in the U.S. Federal Highway Administration NGSIM dataset (U.S. Department of Transportation, Federal Highway Administration, 2016). The position `Local_Y` is converted from feet to meters. The observations are sampled at 10 Hz. Episode extraction requires consecutive 100-ms time steps and uses historical quality-control rules, with future evaluation data excluded from model fitting. Vehicle identifiers can recur across capture periods.

Our principal real-data study evaluates 200 preselected episodes, 100 per site, associated with 113 distinct (site, numerical vehicle-ID) pairs across five coarse capture-time blocks. Each episode uses a single chronological origin with 80 previous position measurements (8 s) and holds out 5, 10 or 20 subsequent frames for 0.5-, 1- and 2-second evaluation. All reported comparisons use the same episode and horizon targets. The empirical outcome is future *recorded position*, not the unobserved noise-free path or independently measured physical jerk.

The selection is a convenience sample from bounded official data queries, not a random sample of all vehicle motions. Correlation within recording scenes, shared time blocks, repeated reported IDs and forecast horizons rules out naive independent-episode population significance tests. Complete source-page hashes, episode catalogs, selected positions and error records are preserved in Experiment004; Experiment006 reuses the exact same real episodes.

5.2. Comparison with smoothing splines, Savitzky–Golay and polynomial trends

Table 3 shows that the GCV-selected smoothing spline attains the lowest pooled forecast RMSE at all three horizons. At half a second, zero-reference PLS with profile likelihood and three-point continuation is relatively close (0.1433 m versus 0.1354 m), but its error increases sharply with the horizon. The local-25 continuation may reduce longer-horizon PLS error at the cost of worse short-horizon behavior. The smoothing penalty and continuation rule must therefore be distinguished.

A secondary fixed-design study with the same 200 episodes uses unknown-μ joint Gaussian REML. Its one-second RMSE is 0.5331 m for three-point continuation and 0.5775 m for local-25 continuation, again above the spline's 0.3480 m. The joint selector chooses the zero-penalty boundary in 161 of 200 episodes, reflecting sensitivity of the Gaussian model to real positions and temporal error structure, not proof that the observations have no noise.

5.3. Illustrative trajectories and an earlier pilot

The original exploratory US-101 pilot uses 15 overlapping chronological origins from five segments associated with only two distinct reported vehicle IDs. Its figures remain useful as *illustrations*; they cannot supersede the larger episode comparison.

For the 15-origin pilot the zero-reference PLS recurrence has the smallest half-second position RMSE among the compared variants (0.0745 m). The GCV-selected natural cubic spline, continued by quadratic Taylor expansion, has the lowest two-second RMSE (0.8085 m against 1.2998 m for the zero-reference PLS recurrence). At one second the two are almost tied (0.2951 versus 0.2958 m). The sample-derived nonzero reference helped the 25-point continuation but not the last-three-point rule. These outcomes illustrate forecast-rule dependence in a very small set of recorded tracks.

The same pilot compared historical CV, GCV, AICc, constrained BIC and profile Gaussian likelihood across difference orders $d = 1, \ldots, 4$. Figure 10 visualizes the resulting criterion-level variation without treating the five criteria

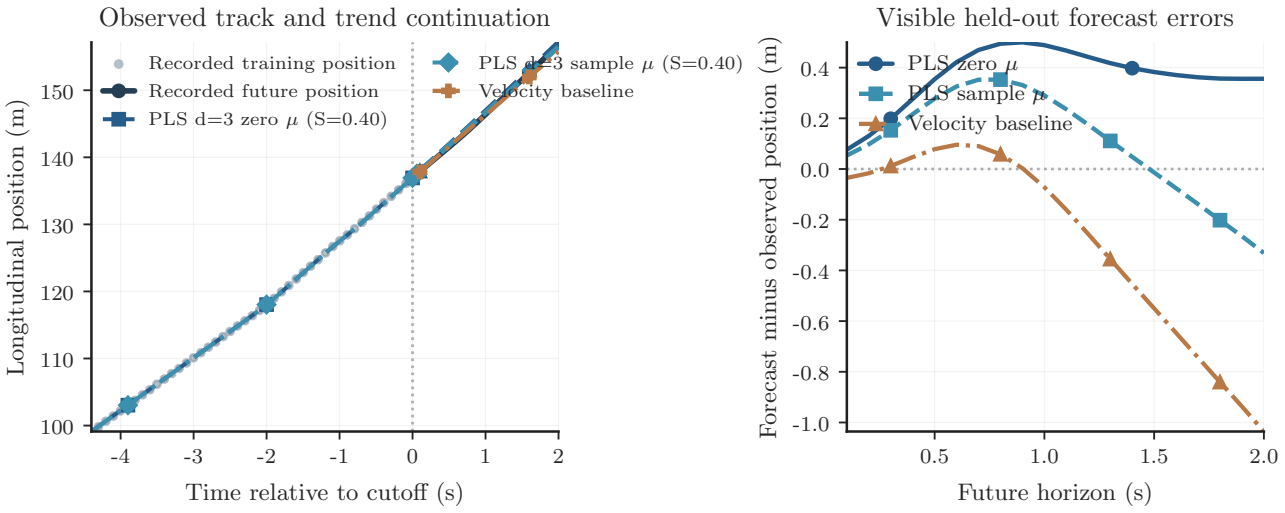


Figure 7: One predeclared US-101 historical training cutoff, fitted $d = 3$ PLS trend and local-polynomial position continuation. Future camera-recorded positions are withheld until evaluation. This is an illustrative pilot segment, not an average over the 200-episode study.

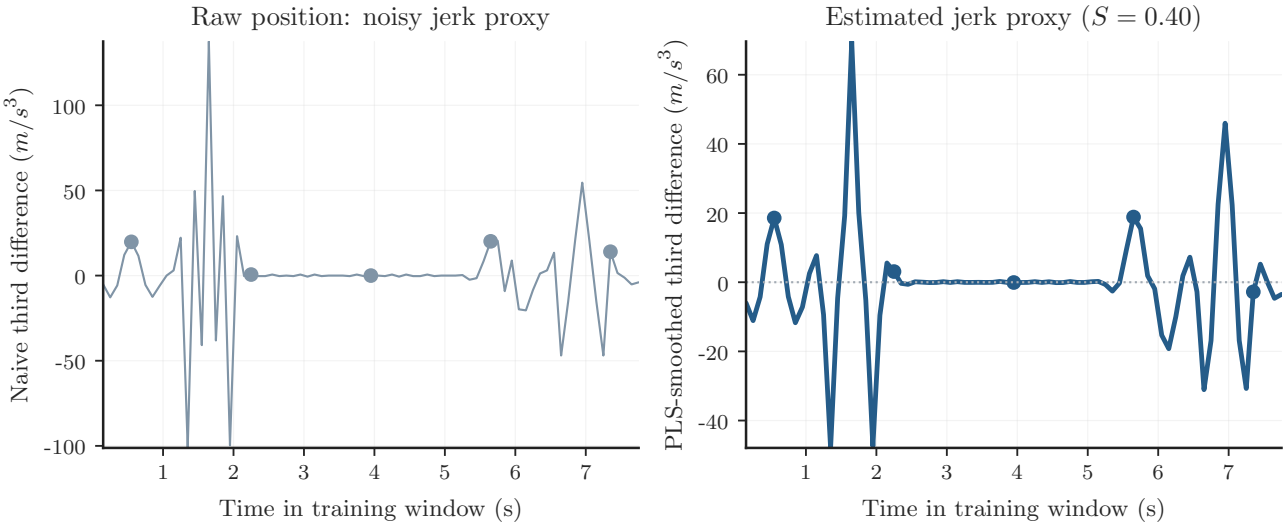


Figure 8: Raw and PLS-smoothed third-difference *jerk proxies* derived from camera positions on one US-101 segment. Panel scales differ and neither plotted derivative is independently known to equal the true physical jerk.

as independent evidence. The full pilot tables and forecast rows remain in the original archive.

6. Discussion and conclusions

This study examines a *statistical smoothing methodology* and its physical interpretation, rather than an algorithmic speed comparison. Third-order penalized least squares (PLS) places an explicit penalty on deviations of discrete jerk from a chosen reference. The normalized smoothing coordinate $S \in [0, 1]$ has an exact relationship to conditional effective degrees of freedom and allows regularization strength to be described without relying on the arbitrary magnitude of λ. This remains useful even where PLS is not the smallest-RMSE method.

6.1. Statistical meaning and identifiability

Under the independent Gaussian variance-ratio model, S is the average spectral fraction of the *third-difference covariance* attributable to observation error, not the proportion of erroneous position observations. The controlled factorial simulation confirms that the joint-REML-selected index can be well calibrated to this theoretical fraction when the assumptions hold: Spearman association 0.965 and mean absolute calibration error 0.0115 in the independently Gaussian condition.

We also varied two quantities that should not be conflated. Increasing measurement noise with true jerk volatility fixed raises S; increasing genuine jerk innovation volatility with measurement noise fixed lowers it. Increasing both

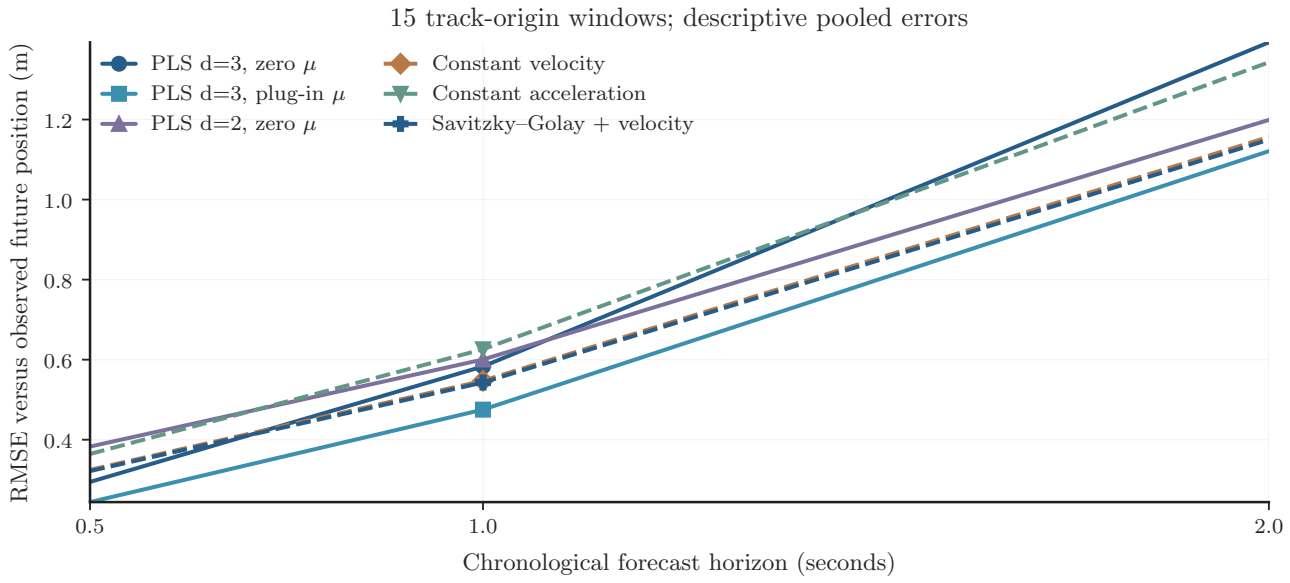


Figure 9: The original exploratory 15-origin pilot: recorded future-position error at three horizons for selected PLS difference orders and conventional constant-velocity, constant-acceleration and Savitzky–Golay benchmarks. Origins overlap within only five recorded segments.



Figure 10: Exploratory one-second recorded-position RMSE on the five-segment pilot, varying PLS difference order, smoothing criterion and training-only reference mean. Colors encode log-RMSE; restricted BIC uses its documented $S \geq 0.10$ bound. This is a descriptive model comparison, not a numerical solver benchmark or significance test.

scales proportionally leaves S unchanged, even when absolute observation-noise variance changes sixteenfold. Consequently, the index is a model-conditional *relative noise-attribution measure*, not a universal indicator of camera accuracy or absolute measurement precision. Serially correlated observation errors induce substantial calibration bias; these failures must accompany the favorable model-correct results.

6.2. Recovery of jerk is distinct from forecasting

A second simulation family contains known analytic motion and its true discrete jerk. In that family the global cubic OLS reference with profiled Gaussian likelihood performed well for historical jerk recovery. Yet the preferred reference policy changes when evaluating future *clean position*: fixed $\mu = 0$ with continuous profile likelihood had lower one-second forecast RMSE among the tested PLS variants. Properly mean-adjusted joint REML optimizes a likelihood, not the jerk-reconstruction or forecasting risk, and was not uniformly best under either loss.

These differences do not undermine the interpretability of S. They demonstrate that smoothness selection, estimation of the reference third difference, and boundary continuation are distinct components of the statistical model. A local 25-point polynomial continuation and a last-three-point recurrence can forecast very differently from the same fitted trend. Neither is a complete probabilistic forecast model for a driver whose acceleration changes unexpectedly.

6.3. What the recorded NGSIM comparison establishes

The 200-episode real-data application is an important counterweight to favorable synthetic outcomes. For the recorded-future position criterion, a GCV-selected natural cubic smoothing spline with quadratic Taylor continuation has the lowest pooled RMSE at all examined horizons. The PLS three-point recurrence is relatively close at 0.5 s, but its longer-horizon error is larger. Joint REML selects the zero-smoothing boundary in 161 of 200 real episodes, highlighting a sensitivity to Gaussian noise assumptions and boundary selection, not an assertion that the camera measurements are noise free.

A separate exploratory five-segment pilot occasionally favors a particular PLS extrapolator, but its 15 origins overlap and represent only two reported vehicle IDs. It cannot outweigh the broader comparison. Moreover, the larger 200-episode sample comes from only five coarse recording-time blocks, and repeated reported IDs need not denote the same physical driver. Neither convenience sample supports robust population-level claims of statistical superiority. The real data contain no independently observed true jerk and thus cannot validate physical jerk reconstruction directly.

6.4. Scope of the contribution

The underlying PLS estimator, difference penalties, and percentage smoothness have important historical precedents in the work of Guerrero (2007, 2008) and subsequent comparisons by Cortés-Toto et al. (2017). We do not claim to have invented PLS smoothing or general PLS forecasting. The applied contribution here is to combine a third-difference kinematic interpretation, an explicit normalized smoothness coordinate, reference-jerk policies, known-truth calibration and recovery experiments, and a real NGSIM comparison against established smoothing methods.

Further investigation should use independently documented measurement noise or multiple sensors, more independent recording sessions, and covariance models allowing serial correlation. A dynamic physical forecasting model is also necessary before making broad claims about future maneuver prediction.

Reproducibility and data rights

The U.S. DOT/FHWA NGSIM dataset (DOI: 10.21949/1504477) supplies the observed positions; the dataset portal identifies a CC BY-SA 3.0 license. Source queries and original page hashes, episode catalogs, all forecasts and model-selection rows, and the distinction between recorded and simulated truth are documented in the case-study repository. The two simulation studies preserve every generated clean position, noisy observation, true discrete jerk, reference-mean policy, fit error and selected S. All source code, simulation configurations, first-party data, bootstrap summaries, vector figures, and run-specific SHA-256 inventories are retained in the Experiment004–007 and manuscript artifact archives. No underlying NGSIM observation was artificially altered to create purported real ground-truth jerk.

Declarations

This is a research manuscript draft. Authorship, correspondence, funding, competing interests and any journal submission require coauthor review and agreement.

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